

Day 18 – Brute-Force Attacks

What is a Brute-Force Attack? (Simple)

A brute-force attack is when:

An attacker repeatedly tries many password combinations until one works.

This succeeds **only when**:

- Passwords are weak
- Login attempts are unlimited
- No detection or blocking exist

Brute Force Attacks

Brute Force Attacks Explained

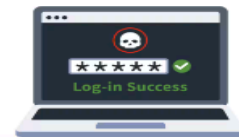
In a brute force attack, a cybercriminal uses trial and error to try and break into a device, network, or website.



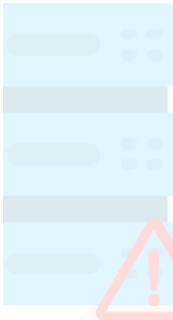
An attacker utilizes a hacking tool.



The hacking tool attempts multiple logins.



The system returns a valid or invalid response.



Attacker

Server

Example Scenario

Target : An online shopping website login page.

Victim Account

Email: user@gmail.com

Password: password123

 This password is weak and common.

Why the Attack Works (Core Reasons)

✗ Security Mistakes by Website

- No limit on login attempts
- No CAPTCHA
- No account lockout
- No alert on multiple failures

🔄 What Happens During the “Simulation”

Step 1: Repeated Login Attempts

The system receives:

```
Attempt 1 → Failed
Attempt 2 → Failed
Attempt 3 → Failed
...
Attempt 50 → Success
```

📌 The website **never stops** or slows down attempts.

What the SYSTEM LOGS Look Like (Real Evidence)

Web Server Log Example

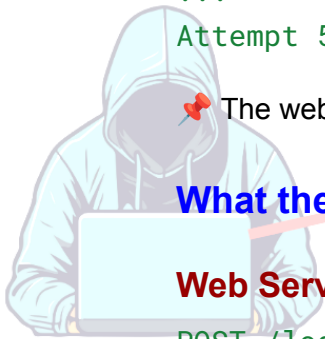
```
POST /login 401
POST /login 401
POST /login 401
POST /login 200
```

Meaning:

- 401 → login failed
- 200 → login successful

📌 This sudden success after many failures is a **red flag**.

Brute Force Attacks



Attacker

Automated Tool

Server